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† Not required for the remainder of the text.

‡ This section is a prerequisite for Sections 17 and 36 only.

Instructor's Preface

This is an introduction to abstract algebra. It is anticipated that the students have studied calculus and probably linear algebra. However, these are primarily *mathematical maturity* prerequisites; subject matter from calculus and linear algebra appears mostly in illustrative examples and exercises.

As in previous editions of the text, my aim remains to teach students as much about groups, rings, and fields as I can in a first course. For many students, abstract algebra is their first extended exposure to an axiomatic treatment of mathematics. Recognizing this, I have included extensive explanations concerning what we are trying to accomplish, how we are trying to do it, and why we choose these methods. Mastery of this text constitutes a firm foundation for more specialized work in algebra, and also provides valuable experience for any further axiomatic study of mathematics.

Changes from the Sixth Edition

The amount of preliminary material had increased from one lesson in the first edition to four lessons in the sixth edition. My personal preference is to spend less time before getting to algebra; therefore, I spend little time on preliminaries. Much of it is review for many students, and spending four lessons on it may result in their not allowing sufficient time in their schedules to handle the course when new material arises. Accordingly, in this edition, I have reverted to just one preliminary lesson on sets and relations, leaving other topics to be reviewed when needed. A summary of matrices now appears in the Appendix.

The first two editions consisted of short, consecutively numbered sections, many of which could be covered in a single lesson. I have reverted to that design to avoid the cumbersome and intimidating triple numbering of definitions, theorems examples, etc. In response to suggestions by reviewers, the order of presentation has been changed so

that the basic material on groups, rings, and fields that would normally be covered in a one-semester course appears first, before the more-advanced group theory. Section 1 is a new introduction, attempting to provide some feeling for the nature of the study.

In response to several requests, I have included the material on homology groups in topology that appeared in the first two editions. Computation of homology groups strengthens students' understanding of factor groups. The material is easily accessible; after Sections 0 through 15, one need only read about free abelian groups, in Section 38 through Theorem 38.5, as preparation. To make room for the homology groups, I have omitted the discussion of automata, binary linear codes, and additional algebraic structures that appeared in the sixth edition.

I have also included a few exercises asking students to give a one- or two-sentence synopsis of a proof in the text. Before the first such exercise, I give an example to show what I expect.

Some Features Retained

I continue to break down most exercise sets into parts consisting of computations, concepts, and theory. Answers to odd-numbered exercises not requesting a proof again appear at the back of the text. However, in response to suggestions, I am supplying the answers to parts a), c), e), g), and i) only of my 10-part true–false exercises.

The excellent historical notes by Victor Katz are, of course, retained. Also, a manual containing complete solutions for all the exercises, including solutions asking for proofs, is available for the instructor from the publisher.

A dependence chart with section numbers appears in the front matter as an aid in making a syllabus.

Acknowledgments

I am very grateful to those who have reviewed the text or who have sent me suggestions and corrections. I am especially indebted to George M. Bergman, who used the sixth edition and made note of typographical and other errors, which he sent to me along with a great many other valuable suggestions for improvement. I really appreciate this voluntary review, which must have involved a large expenditure of time on his part.

I also wish to express my appreciation to William Hoffman, Julie LaChance, and Cindy Cody of Addison-Wesley for their help with this project. Finally, I was most fortunate to have John Probst and the staff at TechBooks handling the production of the text from my manuscript. They produced the most error-free pages I have experienced, and courteously helped me with a technical problem I had while preparing the solutions manual.

Suggestions for New Instructors of Algebra

Those who have taught algebra several times have discovered the difficulties and developed their own solutions. The comments I make here are not for them.

This course is an abrupt change from the typical undergraduate calculus for the students. A graduate-style lecture presentation, writing out definitions and proofs on the board for most of the class time, will not work with most students. I have found it best

to spend at least the first half of each class period answering questions on homework, trying to get a volunteer to give a proof requested in an exercise, and generally checking to see if they seem to understand the material assigned for that class. Typically, I spent only about the last 20 minutes of my 50-minute time talking about new ideas for the next class, and giving at least one proof. From a practical point of view, it is a waste of time to try to write on the board all the definitions and proofs. They are in the text.

I suggest that at least half of the assigned exercises consist of the computational ones. Students are used to doing computations in calculus. Although there are many exercises asking for proofs that we would love to assign, I recommend that you assign at most two or three such exercises, and try to get someone to explain how each proof is performed in the next class. I do think students should be asked to do at least one proof in each assignment.

Students face a barrage of definitions and theorems, something they have never encountered before. They are not used to mastering this type of material. Grades on tests that seem reasonable to us, requesting a few definitions and proofs, are apt to be low and depressing for most students. My recommendation for handling this problem appears in my article, *Happy Abstract Algebra Classes,* in the November 2001 issue of the *MAA FOCUS.*

At URI, we have only a single semester undergraduate course in abstract algebra. Our semesters are quite short, consisting of about 42 50-minute classes. When I taught the course, I gave three 50-minute tests in class, leaving about 38 classes for which the student was given an assignment. I always covered the material in Sections 0–11, 13–15, 18–23, 26, 27, and 29–32, which is a total of 27 sections. Of course, I spent more than one class on several of the sections, but I usually had time to cover about two more; sometimes I included Sections 16 and 17. (There is no point in doing Section 16 unless you do Section 17, or will be doing Section 36 later.) I often covered Section 25, and sometimes Section 12 (see the Dependence Chart). The job is to keep students from becoming discouraged in the first few weeks of the course.

Student's Preface

This course may well require a different approach than those you used in previous mathematics courses. You may have become accustomed to working a homework problem by turning back in the text to find a similar problem, and then just changing some numbers. That may work with a few problems in this text, but it will not work for most of them. This is a subject in which understanding becomes all important, and where problems should not be tackled without first studying the text.

Let me make some suggestions on studying the text. Notice that the text bristles with definitions, theorems, corollaries, and examples. The definitions are crucial. We must agree on terminology to make any progress. Sometimes a definition is followed by an example that illustrates the concept. Examples are probably the most important aids in studying the text. *Pay attention to the examples.* I suggest you skip the proofs of the theorems on your first reading of a section, unless you are really "gung-ho" on proofs. You should read the statement of the theorem and try to understand just what it means. Often, a theorem is followed by an example that illustrates it, a great aid in really understanding what the theorem says.

In summary, on your first reading of a section, I suggest you concentrate on what information the section gives, and on gaining a real understanding of it. If you do not understand what the statement of a theorem means, it will probably be meaningless for you to read the proof.

Proofs are very basic to mathematics. After you feel you understand the information given in a section, you should read and try to understand at least some of the proofs. Proofs of corollaries are usually the easiest ones, for they often follow very directly from the theorem. Quite a lot of the exercises under the "Theory" heading ask for a proof. Try not to be discouraged at the outset. It takes a bit of practice and experience. Proofs in algebra can be more difficult than proofs in geometry and calculus, for there are usually no suggestive pictures that you can draw. Often, a proof falls out easily if you happen to

look at just the right expression. Of course, it is hopeless to devise a proof if you do not really understand what it is that you are trying to prove. For example, if an exercise asks you to show that given thing is a member of a certain set, you must *know* the defining criterion to be a member of that set, and then show that your given thing satisfies that criterion.

There are several aids for your study at the back of the text. Of course, you will discover the answers to odd-numbered problems not requesting a proof. If you run into a notation such as $\mathbb{Z}_n$ that you do not understand, look in the list of notations that appears after the bibliography. If you run into terminology like *inner automorphism* that you do not understand, look in the Index for the first page where the term occurs.

In summary, although an understanding of the subject is important in every mathematics course, it is really crucial to your performance in this course. May you find it a rewarding experience.

Narragansett, RI *J.B.F.*

Dependence Chart

